Fisher forecasts from cached derivatives

How the derivatives are computed and stored, and why every FoM, error and shift is recomputed in seconds

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1 Summary

The question. The project's Fisher notebooks report figures of merit (FoMs), marginalized errors and parameter shifts for four bias variants, up to four linear-bias priors, seven scale cuts and four bin layouts, under dozens of assumed truths. One likelihood evaluation runs CAMB and CosmoLike's non-Limber projection of every thin bin, and the derivatives of one layout took roughly 15 to 45 minutes during development. How, then, can a change of model, prior or scale cut cost only seconds?

The answer.

1. A Fisher forecast linearizes the data vector around one point, so it needs the derivatives at that point and nothing else (Section 2).

2. For each layout the notebooks compute a fixed set of base columns once, the derivatives with respect to the cosmology, to four bias operators in every bin and to the terms linear in their amplitude, and cache them on disk (Section 3).

3. A derivative needs more accuracy than a χ^2: numerical noise divided by the step looks like information and inflates the FoM. A convergence study sets the final accuracy tier, the one every published number uses (Section 4).

4. A bias model is a fixed matrix M that combines the base columns, $\mathsf{D} = \mathsf{BM}$, by the chain rule. A new model is a new M (Section 5).

5. A scale cut keeps a subset of the rows, and each cut is reduced once to the matrix $\mathsf{G} = \mathsf{B}^\mathsf{T}\mathsf{C}^{-1}\mathsf{B}$, at most 645×645 (Section 6). Every Fisher matrix is then $\mathsf{M}^\mathsf{T}\mathsf{GM}$ plus priors, and every FoM, shift and $\langle\chi^2\rangle$ is a product of small matrices (Section 7).

6. New likelihood evaluations are needed only for a term outside the base columns, a new fiducial point, or effects beyond first order (Section 8). Section 9 lists the cache files and what invalidates them.

The setup. The data are the DESI band powers C_ℓ^{gg}, the angular power spectrum averaged in 53 multipole bands, computed without the Limber approximation for six tracer groups g (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four layouts: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$. The bias models A and B and the linear-bias priors are those of the correlated-prior note, `notes/correlated_bias_prior_v2`.

2 Why derivatives at one point are enough

Let $m(p)$ be the expected data vector at the parameters p, p_0 the fiducial point and C the covariance of the data. The vector m has n entries: the band powers of every bin's auto spectrum and of every measured cross pair of bins, after 795 entries that the data format reserves for cosmic-shear blocks and that stay zero for DESI alone. Near p_0

$$m(p) \;\simeq\; m(p_0) + \mathsf{D}\,(p - p_0), \qquad \mathsf{D}_{aj} = \left.\frac{\partial m_a}{\partial p_j}\right|_{p_0}, \tag{1}$$

so a Gaussian likelihood becomes a Gaussian in p. With Gaussian priors of precision $F_{\rm prior}$, the posterior has the inverse covariance

$$F \;=\; \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D} \;+\; F_{\rm prior}, \tag{2}$$

the Fisher matrix. Its inverse is the forecast: $\sigma_j = [(F^{-1})_{jj}]^{1/2}$ is the marginalized error of parameter j, and $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$, with C_w the (w_0, w_a) block of F^{-1} and $w(a) = w_0 + w_a(1-a)$ the dark-energy equation of state (a the scale factor). $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way, with Ω_m the matter density today in units of the critical density and A_s the amplitude of the primordial curvature power spectrum.

The likelihood enters F only through D, at one point. The covariance is Gaussian, computed at the one-loop fiducial for 14,000 square degrees, with its inverse rescaled by $f_A = 12{,}300/14{,}000$ to the survey area. It does not depend on the model being fitted.

3 The base columns

For each layout the notebooks compute the columns of a base matrix B: derivatives of m at the fiducial point with respect to a fixed list of quantities, from which the derivatives of every model are built (Section 5). They come in three kinds.

3.1 Stencil derivatives

Twenty columns are derivatives with respect to sampled parameters of the model A likelihood: the seven cosmological parameters ($10^9 A_s$, the spectral index n_s, the Hubble constant H_0, the baryon density Ω_b, Ω_m, w_0 and w_a), the six group amplitudes B_g, and the seven velocity dispersions σ_v that smooth the bins' redshift distributions (the Fingers-of-God effect). B_g sets the linear bias of every bin of group g at the fiducial point, $b_{1,i} = B_g / D(\bar{z}_i)$. Each column is a five-point stencil,

$$\frac{\partial m}{\partial \theta} \;\simeq\; \frac{m(\theta_0 - 2\epsilon) - 8\,m(\theta_0 - \epsilon) + 8\,m(\theta_0 + \epsilon) - m(\theta_0 + 2\epsilon)}{12\,\epsilon}, \tag{3}$$

with every other parameter at its fiducial value. The weights cancel the Taylor terms through fourth order, so the truncation error falls as ϵ^4, against ϵ^2 for a central difference, and vanishes for any dependence up to quartic. The center point has zero weight, so a column costs four evaluations. The step is relative, $\epsilon = h\,\theta_0$; a parameter whose fiducial value is zero (w_a) takes an absolute step, scaled by its prior width when its prior is Gaussian.

Each evaluation is a full call of the likelihood, CAMB included (`ModelFisher.dv` in `interface/cosmolike_lssty1xdesi_fisher.py`, with cobaya's cache disabled so that no

stale vector is reused). The twenty columns cost 80 evaluations. In the model A and B notebooks the six B_g columns enter no Fisher matrix, because the per-bin columns below replace them; they are kept for the chain-rule check of Section 8.

3.2 Per-bin operator columns

In model A the real-space galaxy power spectrum of bin i is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\, P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_{0,i}\, k^2 P_{\rm mm}(k), \tag{4}$$

plus the stochastic terms of Section 3.3. Here $P_{\rm mm}$ is the nonlinear matter power spectrum, $P_{\rm loop}$ the one-loop terms of the quadratic bias b_2 and the tidal bias b_{s^2}, and α_0 the k^2 counterterm, in $h^{-2}\,{\rm Mpc}^2$. The models need the derivatives with respect to these four operators in every bin, each at fixed values of the other three: $\partial m/\partial b_{1,i}$, $\partial m/\partial b_{2,i}$, $\partial m/\partial b_{s^2,i}$ and $\partial m/\partial \alpha_{0,i}$.

Why the setter is intercepted. The sampled parameters offer no per-bin handle for these steps: the parameter files tie each bin's linear bias to its group amplitude, and the likelihood computes b_2 and b_{s^2} from b_1 through calibrations. The likelihood also sets every nuisance value at every point, so a value written into CosmoLike beforehand would be overwritten. The four per-bin values exist separately only in the call through which the likelihood hands them to CosmoLike, `set_nuisance_bias_tidal` of the compiled interface. The notebook replaces that function by a wrapper that adds a step to one operator in a chosen set of bins and passes the values on, and restores the original after the steps. CosmoLike's k^2 term is $2\,b_1 b_K\, k^2 P_{\rm mm}$, with b_K its per-bin coefficient and k in units of H_0/c, so an α_0 step enters as $b_K = \alpha_0/[2\,b_1 (c/H_0)^2]$.

One set of evaluations per color. A spectrum C_ℓ^{ij} depends on the biases of bins i and j only. Shift one operator in every bin of a set S at once. If no measured cross pair joins two bins of S, each spectrum contains at most one shifted bin, so the change of the spectra of bin i (its auto spectrum and the cross pairs that contain it) is caused by bin i alone. One set of stencil evaluations then yields the column of every bin of S: the stencil difference on bin i's spectra, and zero elsewhere, exactly, since bin i's biases enter no other spectrum.

Sets with this property are the colors of a coloring of the pair graph, whose vertices are the bins and whose edges are the measured cross pairs. The notebooks color it greedily (`pair_coloring`): each bin, in order, takes the smallest color that none of its already colored neighbors has. Every layout gets four colors. On $\Delta z = 0.2$ the two LRG3 bins and the two ELG1 bins overlap in redshift and are all paired with each other, so no coloring can use fewer.

The five-point stencil per color. Each operator is stepped by $\pm\epsilon$ and $\pm 2\epsilon$ in every bin of one color at once (a relative step for b_1, absolute steps for b_2, b_{s^2} and α_0), and Eq. (3) combines the four vectors. The four operators then cost $4 \times 4 \times 4 = 64$ evaluations on every layout, whatever the number of bins; bin by bin they would cost 1,216 on $\Delta d_{\rm com} = 62$ Mpc. The steps and the accuracy of these evaluations are the subject of Section 4.

3.3 Exact templates

Three terms of the models are linear in their amplitude, so their derivative is the term itself per unit amplitude, with no truncation error: the white noise beyond Poisson SN_0, in units of

layout	bins N	cross pairs	entries n	$n_{\rm base}$	kept, shipped mask
$\Delta d_{\rm com} = 62$ Mpc	76	97	9,964	547	5,309
$\Delta z = 0.02$	90	133	12,614	645	7,621
$\Delta z = 0.05$	36	52	5,459	267	2,982
$\Delta z = 0.2$	10	13	2,014	85	744

Table 1: The sizes of one layout's Fisher problem. n: entries of the data vector; $n_{\rm base} = 7N+15$: columns of the base matrix; last column: the band powers the shipped mask keeps, the largest covariance block any cut uses.

the bin's Poisson power $1/\bar{n}_i$ ($\bar{n}_i$ its tracer density); the stochastic term $\mathrm{SN}_2\, k^2\mu^2$, with μ the cosine of the angle to the line of sight, in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$ ($f_{\rm sat}$ the satellite fraction, $\sigma_{\rm sat}$ the satellites' velocity dispersion in distance units); and the radial counterterm α_2 of model B, in $h^{-2}\,\mathrm{Mpc}^2$, which enters the counterterm density $\frac{1}{2}(b_1\alpha_0 + f\mu^2\alpha_2)\, k^2\delta_{\rm L}$ (f the linear growth rate, $\delta_{\rm L}$ the linear density contrast).

The likelihood projects SN_0 and SN_2 directly through each bin's Fingers-of-God-convolved kernel $W_i(\chi)$, with χ the comoving distance (`stochastic_templates` of `likelihood/desi_cl.py`, orders 0 and 2): SN_0 through $\int d\chi\, W_i W_j/\chi^2$, and SN_2 through $\int d\chi\, W_i' W_j'/\chi^2$, because $k^2\mu^2 = k_\parallel^2$ is a second radial derivative (the correlated-prior note, section "Model A"). These templates need only the kernels of one fiducial evaluation. α_2 has its own per-bin setter, `set_nuisance_bias_alpha2`, intercepted the same way; a central difference on each color gives its columns, exact because the term is linear, in eight evaluations.

3.4 What the base matrix holds

B has 14 stencil columns (the cosmology and σ_v), seven blocks of one column per bin (b_1, b_2, b_{s^2}, α_0, α_2, SN_0, SN_2), and one check column (Section 8): with N bins, $n_{\rm base} = 7N + 15$ (Table 1). It takes about 150 likelihood evaluations per layout: 80 for the stencil, 64 for the operators, 8 for α_2 and a few fiducial vectors. During development, at the exploration settings of Section 4, one layout took roughly 15 to 45 minutes, a rough figure and not a benchmark.

4 Derivative accuracy

Why a derivative needs more than a χ^2. Every evaluation of m carries numerical noise: the integration grids, the non-Limber χ grid and CAMB's tolerances change the vector by small amounts that are not smooth in the parameters. A finite difference divides that noise by the step, so a setting that converges a χ^2 (the MCMC criterion is $|\Delta\chi^2| < 0.2$) can leave a derivative noisy. The noise biases the forecast one way. A column $d_j + e_j$, with e_j its noise, gives $F_{jj} = (d_j + e_j)^{\mathsf{T}}\mathsf{C}^{-1}(d_j + e_j)$, whose average exceeds the true value by $\langle e_j^{\mathsf{T}}\mathsf{C}^{-1}e_j\rangle > 0$: noise looks like information, and the FoM comes out too high.

The convergence study, `tests/studies/run_fisher_derivatives.py`, measures this on $\Delta z = 0.05$ for the paper model, model B + tilts on the pessimistic prior. It scores the marginalized errors of w_0, w_a, Ω_m and A_s, both FoMs and the shifts of the break-test truths at $k_{\rm max} = 0.075$ and $0.2\,h/\mathrm{Mpc}$. The MCMC settings raise $\mathrm{FoM}(w_0, w_a)$ by 57% at $0.075\,h/\mathrm{Mpc}$ and by 10% at $0.2\,h/\mathrm{Mpc}$, although their fiducial vector differs from the reference by only

configuration	H_0	Ω_m	w_0	w_a
MCMC settings	2.1×10^{-3}	1.7×10^{-2}	2.5×10^{-2}	1.4×10^{-2}
$g = 64$	2.2×10^{-3}	8.1×10^{-3}	1.1×10^{-2}	4.8×10^{-3}
$g = 128$ (reference)	2.2×10^{-3}	6.5×10^{-3}	4.3×10^{-3}	1.8×10^{-3}
$g = 256$	2.2×10^{-3}	6.1×10^{-3}	2.2×10^{-3}	3.5×10^{-4}

Table 2: The noise of one evaluation along each stencil parameter, in $\sqrt{\chi^2}$ units, on $\Delta z = 0.05$. The rows with a χ-grid factor g keep every other control at the reference.

χ-grid factor g	$k_{\max} = 0.075\,h/\mathrm{Mpc}$	$k_{\max} = 0.2\,h/\mathrm{Mpc}$
32 (MCMC)	+38%	+6.7%
64 (old caches)	+31%	+5.1%
128	+0.9%	+0.3%

Table 3: The offset of FoM(w_0, w_a) of the paper model from its value at $g = 256$, on $\Delta z = 0.05$, every other control at the reference. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

$\Delta\chi^2 = 0.015$. The old central-difference caches gave FoM$(w_0, w_a) = 0.227$ at $0.075\,h/\mathrm{Mpc}$ on this layout, against 0.157 converged.

The reference and the ladder. The reference configuration sets every control high: CosmoLike's `accuracyboost` 4 and `integration_accuracy` 2, the low-rank engine's rank 8 and multipole cap 2000, a χ-grid factor $g = 128$, and CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 and transfer $k_{\max} = 80\,\mathrm{Mpc}^{-1}$. Here g is the product `accuracyboost`×`nonlimber_accuracyboost`, and the non-Limber projection samples the comoving distance at $511\,g + 1$ points; the MCMC value on $\Delta z = 0.05$ is $g = 32$. Each other configuration lowers one control, and g is also raised to 256.

The noise of one evaluation. Through five equally spaced stencil points, the fourth difference $m(-2) - 4m(-1) + 6m(0) - 4m(1) + m(2)$ of a smooth vector is of order ϵ^4 times its fourth derivative, negligible at these steps, while noise independent from point to point gives it $1 + 16 + 36 + 16 + 1 = 70$ times the variance of one point. Whitened by the covariance and divided by $\sqrt{70}$, its norm estimates the noise of one evaluation in $\sqrt{\chi^2}$ units (Table 2). At the MCMC settings the Ω_m, w_0 and w_a columns are seven to twelve times noisier than the H_0 column, which does not change with the grid. The finer χ grid quiets w_0 and w_a most: the w_a noise falls fortyfold from the MCMC settings to $g = 256$.

The χ grid dominates. Table 3 gives the offset of FoM(w_0, w_a) from its value at $g = 256$, every other control at the reference. Swapping single columns between configurations shows where the offset comes from: the w_a column carries it, then w_0 and Ω_m; the per-bin and σ_v columns move the FoM by less than 0.1%.

The criterion: independence of the step. The truncation error of a stencil grows with the step, as ϵ^4 for five points, while the noise it divides falls as $1/\epsilon$. A derivative is converged when neither matters: when the FoMs, the errors and the shifts stay put as the step changes, every step scaled together, and as the stencil changes order. At $g = 128$ the five-point stencil

at $h/2$ raises the FoM by 18%, the noise divided by a smaller step. At $g = 256$ the FoM is flat within about $\pm 1\%$ (FoM$(\Omega_m, 10^9 A_s)$ within $\pm 1.4\%$) from $h/2$ to $3h$, and the three-point stencil agrees with the five-point one. The tolerance on the FoMs is about 1 to 2%; $g = 128$ agrees with $g = 256$ to 0.9% only at $h = 0.02$, so it stays a documented cross-check, and the final tier uses $g = 256$ on this layout.

The CAMB halofit patch. Halofit finds the nonlinear scale by bisection on $\sigma(R)$, the rms linear density contrast smoothed on the scale R; stock CAMB stops at $|\sigma(R) - 1| \leq 10^{-3}$. Where the bisection stops jumps between nearby cosmologies and shifts the nonlinear power spectrum by about 0.3%, a noise that a finite difference divides by its step. Cocoa's installation key `PATCH_CAMB_HALOFIT_TOLERANCE`, on by default, tightens the test to 10^{-7}. The final tier requires it: the builders stop at start unless `halofit.f90` holds the 10^{-7} test and the compiled CAMB library is newer than that file (`check_halofit_patch`).

Two tiers. The project fixes two accuracy tiers, which `derivative_settings` returns per layout.

- **Exploration**: fast settings for quick looks; its FoMs are not converged.
- **Final**: the settings the study converged, for every figure and number of a paper or of a note's final version. The criterion is the stability of the FoMs, the errors and the shifts; runtime is not a criterion.

Both tiers use five-point stencils for every stencil and per-bin column, and exact templates for SN_0, SN_2 and α_2. Each cache records its tier and settings, and a fingerprint (`settings_fingerprint`): the sha256 of the resolved cobaya input, which is the example yaml merged with the likelihood's options and parameter files, the accuracy settings, the priors and the ties between parameters, together with the fiducial point and the tier. Unlike a comparison of file names, it changes when a yaml is edited under the same name and stays the same when a file only moves.

TODO, still being measured.

- The full final settings of each layout: CosmoLike's `accuracyboost` and `integration_accuracy`, the low-rank rank and multipole cap, and CAMB's `AccuracyBoost`, k sampling and transfer $k_{\max}$.
- The step check on $\Delta d_{\rm com} = 62$ Mpc at $g = 256$.
- The comparison with DerivKit (adaptive polynomial fits; Cocoa's `DERIVKIT` keys; `get_ddv_dkit`).
- The agreement of the production route, setter interception (Section 3.2), with the study's route, which unties the per-bin parameters and steps them through the likelihood.

5 A model is a matrix

Every parameter of the models moves the per-bin operators in a fixed linear way at the fiducial point. By the chain rule its derivative is a fixed combination of base columns, and for all parameters together

$$\mathsf{D} \;=\; \mathsf{B}\,\mathsf{M}, \tag{5}$$

with M an $n_{\rm base}\times n_{\rm par}$ matrix of numbers that the notebooks write down without any evaluation (`variant_columns` in the model B notebook; Figure 1). The column of M for each kind of parameter:

- **Cosmology and σ_v**: a unit vector that picks the stencil column.
- **A group offset.** Model A sets $b_{2,i} = b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}$ in every bin i of group g, with $b_2^{\rm L}$ the Lazeyras et al. fit [1], so $\partial m/\partial\Delta_{2,g} = \sum_{i\in g}\partial m/\partial b_{2,i}$: ones on the b_2 entries of the group's bins. The tidal offset $\Delta_{s,g}$ and the group's α_0, α_2, SN_0 and SN_2 are built the same way on their own blocks.
- **A tilt.** The + tilts variants let an offset change linearly across the group, $\Delta_{2,i} = \Delta_{2,g} + t_{2,g}(\bar z_i - \bar z_g)$, with $\bar z_g$ the mean of the group's $\bar z_i$, so $\partial m/\partial t_{2,g} = \sum_{i\in g}(\bar z_i - \bar z_g)\,\partial m/\partial b_{2,i}$: the same entries, weighted by $\bar z_i - \bar z_g$. Tilts exist in groups of more than one bin and have flat priors; SN_0 and SN_2 never tilt.
- **The per-bin amplitude with its calibration slopes.** The linear bias of bin i is parametrized by $\beta_i = b_{1,i}\,D(\bar z_i)$, equal to B_g at the fiducial point. b_2 follows b_1 through $b_2^{\rm L}$, and b_{s^2} through co-evolution, $b_{s^2,i} = -\frac{4}{7}(b_{1,i}-1) + \Delta_{s,g}$, so

$$\frac{\partial m}{\partial\beta_i} \;=\; \frac{1}{D(\bar z_i)}\left[\frac{\partial m}{\partial b_{1,i}} + b_2^{\rm L\,\prime}(b_{1,i})\,\frac{\partial m}{\partial b_{2,i}} - \frac{4}{7}\,\frac{\partial m}{\partial b_{s^2,i}}\right], \tag{6}$$

  with $b_2^{\rm L\,\prime}$ the slope of the fit: three entries, on bin i's b_1, b_2 and b_{s^2} columns.
- **The growth-tied prior**: one amplitude per group, $b_{1,i} = B_g/D(\bar z_i)$, whose column is the sum of the group's β_i columns.
- **Free per bin.** A comparison fit, on the free linear-bias prior, frees every higher-order parameter in every bin: one $\Delta_{2,i}$, $\Delta_{s,i}$, $\alpha_{0,i}$, $\alpha_{2,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ per bin, each a unit vector on its per-bin block, with the Gaussian width w_g that the model gives the group parameter.
- **Free per bin, $\sqrt{N}$-scaled**: the same columns with widths $w_g\sqrt{N_g}$, N_g the number of bins in the group. The mean of N_g independent per-bin values then has the width w_g of the group parameter, so the comparison isolates what the grouping buys.

The four variants of the notebooks are four matrices M on one B: model A (one Δ_2, Δ_s, α_0, SN_0 and SN_2 per group, nothing on the α_2 block), model A + tilts (with the tilts of Δ_2, Δ_s and α_0), model B (adds α_2 per group and widens the offset priors) and model B + tilts (which also tilts α_2). Switching between them, or to any other model built from the same operators, needs no likelihood evaluation.

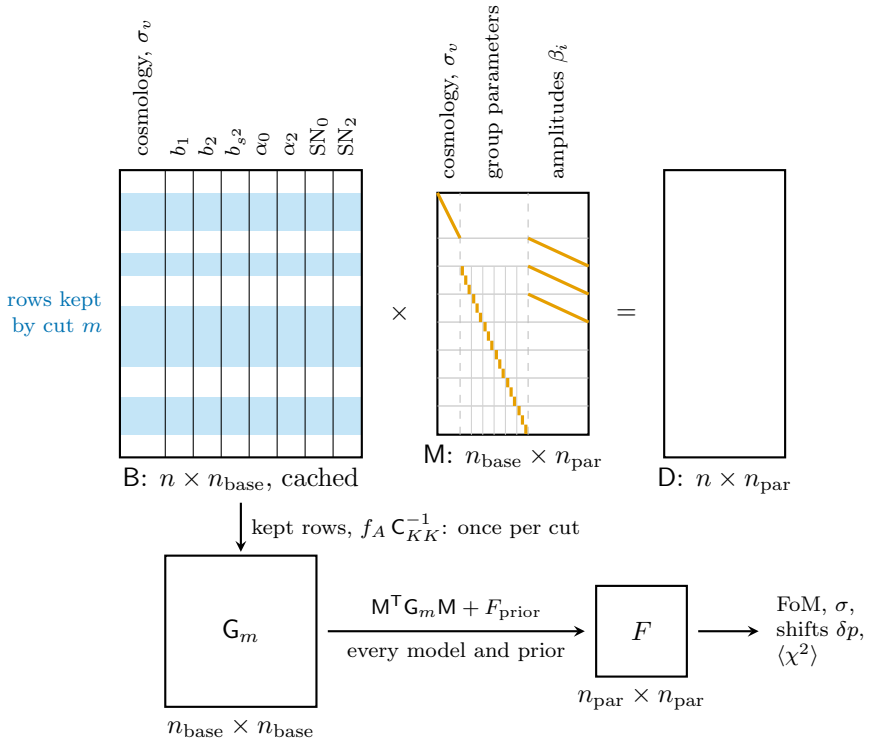


Figure 1: How the notebooks assemble a Fisher matrix. Top: a model is a matrix M on the cached base columns B, and $\mathsf{D} = \mathsf{BM}$ are its derivatives. Orange: the nonzero entries of M, which are the identity on the cosmology and σ_v, one staircase per family of group parameters (ones on the bins of each group; three groups drawn), and three diagonals for the per-bin amplitudes β_i (the chain rule). Bottom: a cut keeps some rows of B (shaded) and is reduced once to G_m; every model and prior at that cut is then $\mathsf{M}^\mathsf{T}\mathsf{G}_m\mathsf{M}$ plus priors. The check column of B is not drawn.

6 A scale cut is a set of rows

A cut at k_{max}, quoted at $z = 0$, cuts bin i at $k_{\text{max}}/D(\bar{z}_i)$ by the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [2], within the shipped mask: a band is kept while less than 5% of it comes from the power spectrum above that wavenumber. The cut changes no derivative; it selects the set K_m of rows that cut m keeps. With B_K the kept rows of B and C_{KK} the kept block of the covariance,

$$\mathsf{G}_m \;=\; f_A\, \mathsf{B}_K^\mathsf{T}\, \mathsf{C}_{KK}^{-1}\, \mathsf{B}_K \tag{7}$$

is an $n_{\text{base}} \times n_{\text{base}}$ matrix that holds everything the data of cut m say about the base columns. The notebooks compute it once per cut (`layout_grams`): they factorize $\mathsf{C}_{KK} = LL^\mathsf{T}$ with L lower triangular (the Cholesky factorization), solve $W = L^{-1}\mathsf{B}_K$ by forward substitution, and form $\mathsf{G}_m = f_A W^\mathsf{T} W$. The inverse covariance is never formed, which saves work and avoids amplifying round-off on an ill-conditioned block, and G_m is symmetric and positive semidefinite by construction. Each layout gets eight such matrices: the seven cuts $k_{\text{max}} = 0.05, 0.075, \ldots, 0.2\,h/\text{Mpc}$, and the shipped mask alone, the reference of the information test

below. With Eq. (5), the data part of the Fisher matrix of any model at any cut is

$$F_{\rm data} \;=\; \mathsf{D}_K^{\mathsf{T}}\, f_A\, \mathsf{C}_{KK}^{-1}\, \mathsf{D}_K \;=\; \mathsf{M}^{\mathsf{T}}\, \mathsf{G}_m\, \mathsf{M}. \tag{8}$$

The model A notebook, which works at one cut, forms $\mathsf{D} = \mathsf{BM}$ explicitly and multiplies it by the inverse of the kept covariance block; the two routes agree (Section 8).

The priors. $F_{\rm prior}$ adds precision and needs no data. The diagonal Gaussian priors, all centered on zero, sit on σ_v (from the example yaml) and on the model's group parameters; the counterterm widths, $0.5/k_{\rm max}^2$, change with the cut, so the diagonal is rebuilt for each cut (`prior_diagonal`). The linear-bias prior acts on the block of the β_i: the baseline and pessimistic priors add the projected precision of the memory model, group by group (`group_bias_precision`; the correlated-prior note, sections "What the prior charges" and "How the prior enters the Fisher matrix"), the free prior adds nothing, and the growth-tied prior is a change of M.

Parameters without data. At small cuts some bins keep no band, and their columns vanish on K_m. A parameter whose information at the cut, data plus prior, falls below 10^{-10} of its information on the shipped mask with the same priors leaves the fit; a Gaussian prior keeps it in. A cut that keeps no band (as $\Delta z = 0.02$ at $k_{\rm max} = 0.05\,h/{\rm Mpc}$) or leaves a cosmological parameter without data gives no fit.

The pseudo-inverse. A plain inverse fails in one more case. On $\Delta d_{\rm com} = 62$ Mpc at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ only 7 of the 18 BGS bins keep a band, and under model A + tilts on the free prior the flat-prior BGS tilts leave one combination of parameters with no information at all. Along such a direction a plain inverse amplifies round-off by about 10^{16}; it once produced a spurious 2σ shift. `stable_inverse` therefore scales F to unit diagonal, $\tilde{F} = SFS$ with $S = {\rm diag}(F)^{-1/2}$, so that the parameters' units do not matter, diagonalizes $\tilde{F} = V\Lambda V^{\mathsf{T}}$, drops the directions whose eigenvalue is below 10^{-12} of the largest, and returns

$$F^{-1} \;=\; S\, V_\kappa\, \Lambda_\kappa^{-1}\, V_\kappa^{\mathsf{T}}\, S, \tag{9}$$

with κ the kept directions. A dropped direction must leave (w_0, w_a), Ω_m and A_s alone: the scan gives no fit when one touches them by more than 10^{-6}.

7 Everything downstream reuses G

Once G_m is built, nothing in the notebooks touches an $n \times n$ matrix or the likelihood again. F is Eq. (8) plus the priors, inverted by Eq. (9), and σ_j and the FoMs follow as in Section 2.

Shifts. A truth that departs from the fitted model moves the data by δm. When δm lies in the span of the base columns, $\delta m = \mathsf{B}c$ with c a vector of $n_{\rm base}$ coefficients: $\alpha_2 = 12.5\,h^{-2}\,{\rm Mpc}^2$ in every bin is $c = 12.5$ on the α_2 block, an offset truth puts its values on the b_2 and b_{s^2} blocks, and a fractional drift δ_i of the linear biases is $c = \sum_i \beta_i\,\delta_i\,\mathsf{M}_{\beta_i}$, with M_{β_i} the column of Eq. (6). To first order the fit moves by

$$\delta p \;=\; F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\delta m \;=\; F^{-1}\mathsf{M}^{\mathsf{T}}\,\mathsf{G}_m\, c \tag{10}$$

(the correlated-prior note, "The shift formula"). $\delta m = \mathsf{B}c$ is exact for the α_2, SN_0 and SN_2 truths, whose terms are linear, and first order for truths in b_1, b_2 and b_{s^2}, since the band powers are quadratic in the biases. A truth costs one product with G_m.

Random truths. Eq. (10) is linear in c, so a family of random truths collapses into one matrix: for drifts δ_i of the linear biases the (w_0, w_a) shift is $\delta w = A\,\delta$, with A the (w_0, w_a) rows of Eq. (10) applied to the N columns $\beta_i \mathsf{M}_{\beta_i}$. Two thousand truths cost one matrix product.

The exact $\langle\chi^2\rangle$. Whether a fit's error bars are honest is tested by averaging $\chi^2_w = (\hat{w} - w_{\mathrm{true}})^{\mathsf{T}} \mathsf{C}_w^{-1} (\hat{w} - w_{\mathrm{true}})$, with $\hat{w}$ the fitted (w_0, w_a), over the truths and the data noise. Every source is Gaussian and enters linearly, so the average is exact in closed form (the correlated-prior note, "The test statistic"):

$$\langle\chi^2_w\rangle \;=\; \Delta^2_{\mathrm{drift}} \;+\; \mathrm{tr}\Big[\mathsf{C}_w^{-1}\left(A\,\Sigma\,A^{\mathsf{T}} + R\,\Sigma_\theta\,R^{\mathsf{T}} + N_w\right)\Big]. \tag{11}$$

Here Δ_{drift} is the (w_0, w_a) shift of the known drift alone, in units of the ellipse; Σ the covariance of the truth's linear-bias residuals; R and Σ_θ the response to, and the prior variance of, any other parameter drawn at random (α_2 per group in the random test of model B); and N_w the (w_0, w_a) block of $F^{-1} F_0 F^{-1}$, the share of the data noise and the Gaussian priors, with F_0 the Fisher matrix without the linear-bias precision. The notebooks report $\sqrt{\langle\chi^2\rangle / n_{\mathrm{dof}}}$, with $n_{\mathrm{dof}} = 2$ for (w_0, w_a), and $n_{\mathrm{dof}} = 1$ for Ω_m or A_s with that parameter's own row; 1 means honest error bars. The random draws only make the percentile bars of the figures.

The cost. Each G_m costs one Cholesky factorization of at most 7,621 rows, about $n_K^3/3 \approx 1.5 \times 10^{11}$ floating-point operations, and one triangular solve, eight times per layout. A Fisher system then costs $\mathsf{M}^{\mathsf{T}}\mathsf{G}_m\mathsf{M}$, with $n_{\mathrm{base}} \le 645$ and $n_{\mathrm{par}} \le 164$ (model B + tilts on $\Delta z = 0.02$), about $2\,n_{\mathrm{base}}^2 n_{\mathrm{par}} \approx 1.4 \times 10^8$ operations, plus the diagonalization of an $n_{\mathrm{par}} \times n_{\mathrm{par}}$ matrix: a fraction of a second. The k_{max} scan of the model B notebook has 336 systems (four layouts, seven cuts, four variants, and the baseline, pessimistic and free priors; the growth-tied prior fails the accuracy test and is left out), 324 with a fit and 12 without any band. In operations, all of them together cost less than the Cholesky factorizations of one layout. Without the decomposition, each variant would need its own derivatives, at the cost of Section 3.4 per layout, and each truth its own data vector.

8 Limits and checks

What needs new likelihood evaluations.

- **A physical term outside the span of B.** A term that is no combination of the per-bin operators and templates, such as a nonzero third-order bias b_3 or redshift-space one-loop terms (which CosmoLike lacks), needs new base columns. A new combination of existing operators, such as a quadratic tilt or another calibration slope at the same fiducial values, is only a new M.
- **A different fiducial point.** Every derivative is taken at p_0. A new fiducial cosmology, new fiducial biases, or a calibration law that moves the fiducial b_2 needs new base columns,

and a new covariance, which sits at the one-loop fiducial. A new covariance alone needs no likelihood evaluation: the notebooks rebuild every G_m from the current covariance file at every run.

- **Dependence beyond first order.** The forecast is the curvature at p_0, and Eq. (10) is first order in δm. The band powers are quadratic in the biases, so a large truth, such as an offset of twice a prior width or the 22% BGS flux-limit drift, is treated to first order, and a shift near the $\sigma/3$ threshold needs a refit, with new evaluations, to be confirmed.

The checks that guard the machinery.

- **The chain rule.** The per-bin columns, obtained through the intercepted setter and summed over a group, $\sum_{i\in g} \partial m/\partial\beta_i$ with Eq. (6), must equal the stencil derivative $\partial m/\partial B_g$, obtained through the full likelihood. The two routes share no evaluation; they agree to 1.2×10^{-7} of the stencil column's largest entry (the correlated-prior note, "Numerical checks").

- **Bitwise template checks.** The order-0 templates of a template run must equal the cached SN_0 templates exactly, and the run's fiducial vector the cached one. B carries the difference of the two vectors as its check column, so their $\Delta\chi^2$ is a diagonal entry of G on the shipped mask; it is zero. After the α_2 steps the vector must equal the one before bit for bit (the setter is back at zero), and the forward and backward α_2 differences must agree (the term is linear).

- **The matched-prior check.** When the truth's linear-bias scatter equals the fit's prior and every other parameter is drawn from the fit's own priors, the projected precision P obeys $\mathsf{P}\Sigma\mathsf{P} = \mathsf{P}$, and the scatter and noise terms of Eq. (11) add up to exactly n_{dof} (the correlated-prior note, "The test statistic"). The notebooks verify this on every system to round-off, which tests G, M, the priors and the pseudo-inverse together. The model A FoMs of the model B notebook's scan also equal those of the model A notebook, assembled by the other route of Section 6, to 7.9×10^{-10}, and a + and a − truth give shifts of equal size.

- **The cache check.** Each cache stores its settings, tier and fingerprint, and a notebook refuses a cache whose entries differ from its own (Section 9).

9 Using the caches

The caches live in `chains/`, which git ignores: a fresh clone recomputes them (hours), and every later run reads them. Their names start with `EXAMPLE_EVALUATE_FISHER`, and $\langle n\rangle = 1$ to 4 numbers the layouts in the order $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2.

- `_SCALE_CUTS.pkl`, written by the linear notebook, `EXAMPLE_EVALUATE_FISHER.ipynb`: the bands each ultraviolet-fraction cut keeps, at $k_{\max} = 0.05, 0.075, \ldots, 0.25\,h/\mathrm{Mpc}$; the scans use the seven cuts up to 0.2.

- `_MODEL_A_DERIVATIVES_<n>.pkl`, written by the model A notebook, `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb`, flag `RECOMPUTE`: the fiducial point and vector, the 20 stencil

columns, the per-bin columns of b_1, b_2, b_{s^2} and α_0, the SN_0 templates, the fiducial per-bin biases, the group of each bin, the Gaussian prior widths, and the settings with their tier and fingerprint.

- `_MODEL_A_SN2_<n>.pkl`, model A notebook, flag `RECOMPUTE_SN2`: the SN_2 templates, kept apart so that the derivative caches stay as they are.
- `_MODEL_B_TEMPLATES_<n>.pkl`, written by the model B notebook, `EXAMPLE_EVALUATE_FISHER_MODEL_B.ipynb`, flag `RECOMPUTE_TEMPLATES`: the α_2 columns and the SN_0 and SN_2 templates.

The model A and B notebooks cache no Fisher matrix, G_m or FoM: these depend on the covariance and are rebuilt at every run.

The flags. With a flag at `False`, the default, a notebook reads an existing cache after the settings check and computes a missing one; `True` recomputes it and overwrites the file. The layouts are computed cheapest first, and each cache is written as soon as its layout is done, so an interrupted run keeps the finished layouts. The model B notebook reads the model A caches and never writes them; it stops when one is missing or was made with other settings. For a fresh clone, run the linear notebook (for the cuts), then the model A notebook, then the model B notebook.

What invalidates a cache.

- **Detected automatically**: any change of the resolved cobaya input that the fingerprint hashes (a likelihood option, a CAMB argument, a prior, a tie between parameters, the fiducial point), of the tier, of the stencil rule or of the steps. The notebook stops and asks for the flag. A file that moves or is renamed with the same content keeps its cache.
- **Not detected, so set the flag by hand**: a change of the code itself (the likelihood's Python, CosmoLike, CAMB beyond the halofit check), since the fingerprint hashes the input, not the code. No check compares a cache with the current code unless a cache is recomputed; the fiducial-vector check then catches a change that moves the fiducial vector.
- **Needs no recomputation**: a new covariance, a new prior or prior width, a new model built from the cached operators, and any cut already in the cut cache. A $k_{\max}$ outside it needs the linear notebook to compute that cut (CosmoLike's split of each band into the power below and above the cut), but no new derivative.

References

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[2] E. Saraivanov, H. Rubira, V. Miranda and T. Eifler, "Towards the Two-Loop EFTofLSS in Galaxy Lensing Surveys", 2026, arXiv:2603.13031.